Satellite Data Analysis Waters Surrounding Taiwan Using Python

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01 - Why choose this topic ?

02 - Where does the data from ?

03 - Steps to analysis data

04 - What can be seen from the results

01 - Why choose this topic ?

Although MATLAB is commonly used for processing satellite data in oceanic and atmospheric sciences, Python offers several advantages:

- Free and open-source: Python is free to use, while MATLAB requires a paid license
- Strong community support: Large and active community makes it easier to find solutions
- Rich scientific libraries: Includes NumPy, SciPy, Pandas, and Matplotlib for data analysis
- Machine learning capability: Widely used for big data processing and machine learning

Despite these advantages, MATLAB remains a powerful tool for scientific computing in engineering.

<GDP Data>

- **Sources: National Bureau of Statistics of China**
- **Taipei Economic and Business Information Network**
- **Year: 2022**

02 - Where does the data from ?

Satellite Data (MODIS-Aqua)

- Source: NASA/GSFC
- Variable: Chlorophyll-a (CHL)
- Product: L3SMI, 4 km resolution

- Time: Nov 25 – Dec 2, 2022
- Region: Waters surrounding Taiwan
- Projection: Equidistant Cylindrical

03 - Steps to analysis data

1. Load satellite data (.nc file) using NetCDF4

2. Extract chlorophyll concentration, latitude, and longitude

3. Convert masked values to NaN for data cleaning

4. Define study region and input city coordinates

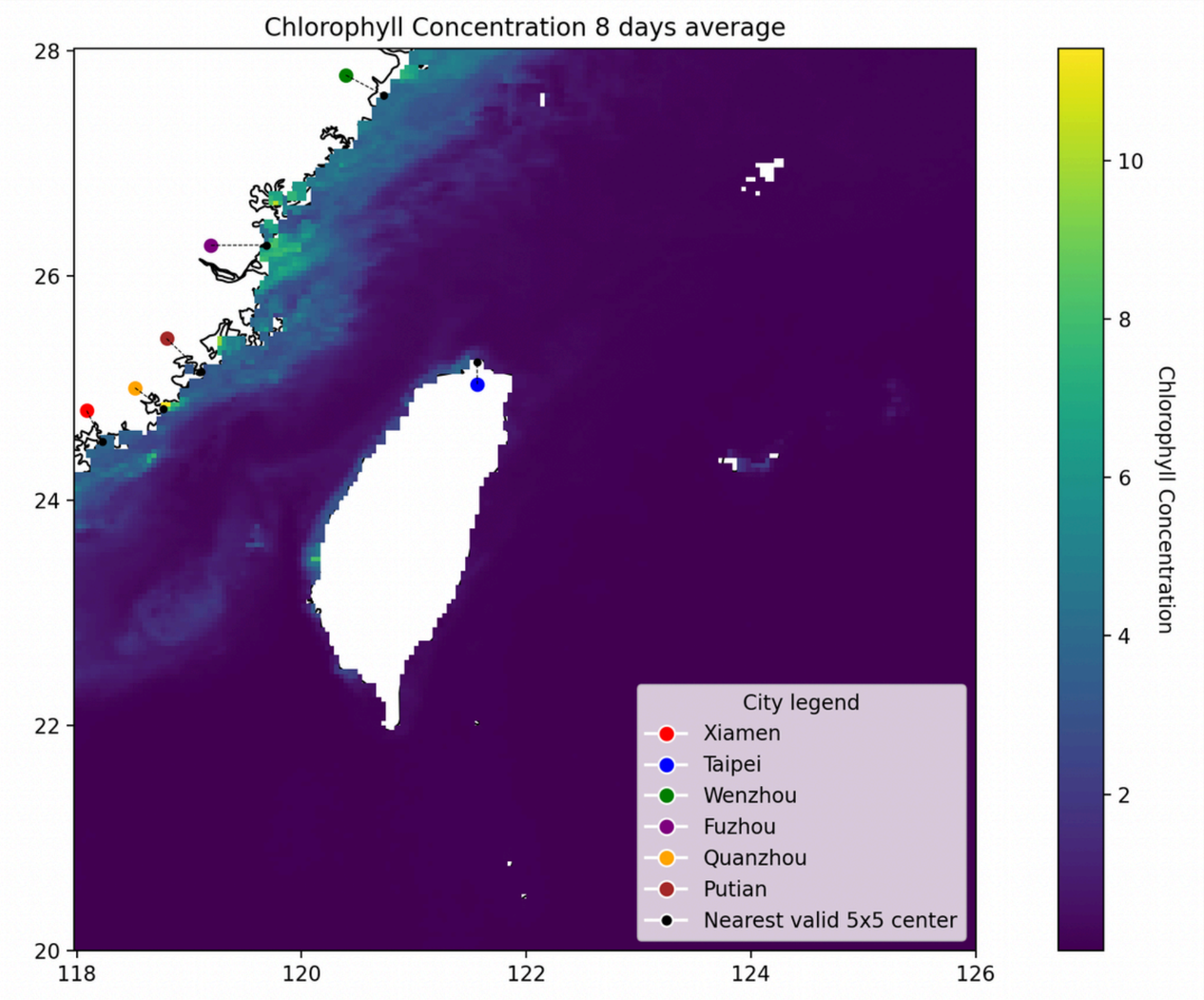
5. Locate nearest candidate grid points based on geographic distance

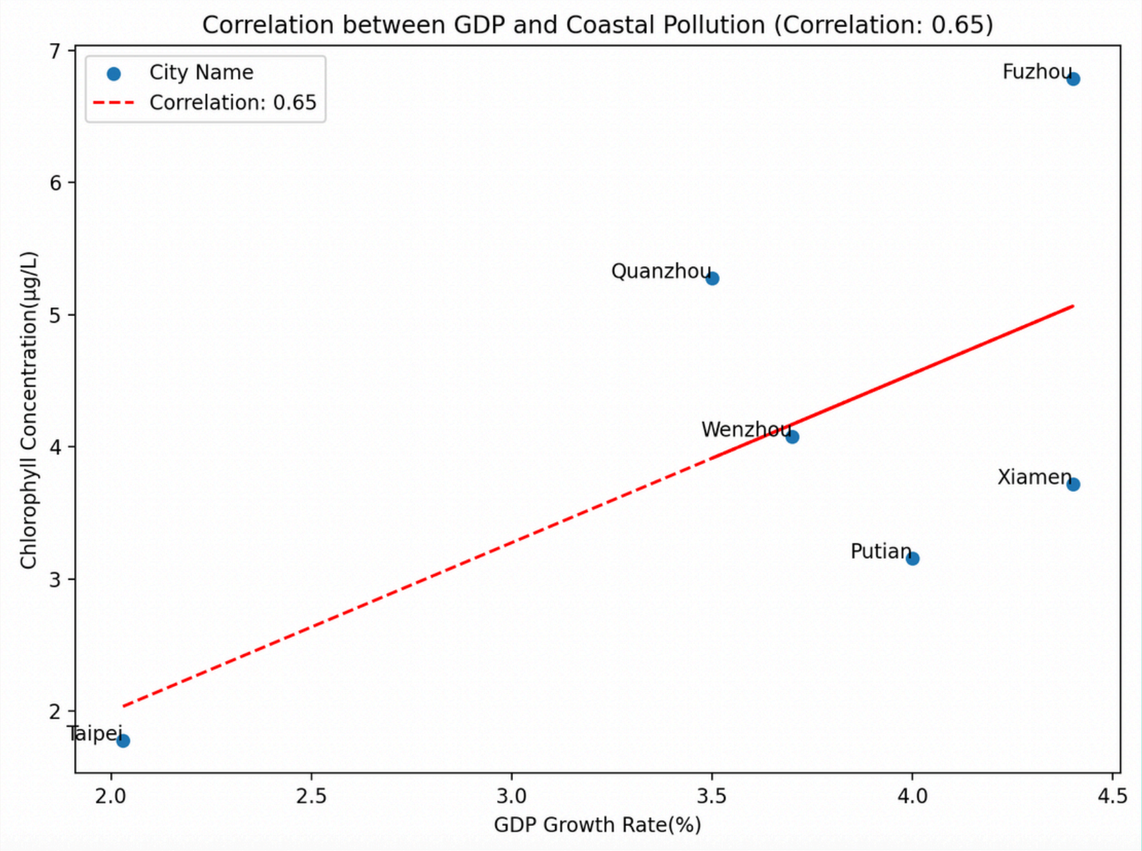
6. Extract 5×5 window around each candidate grid

7. Apply spatial averaging (exclude NaN values, require ≥50% valid data)

8. Visualize chlorophyll distribution and map city locations

04 - What can be seen from the results





Thanks